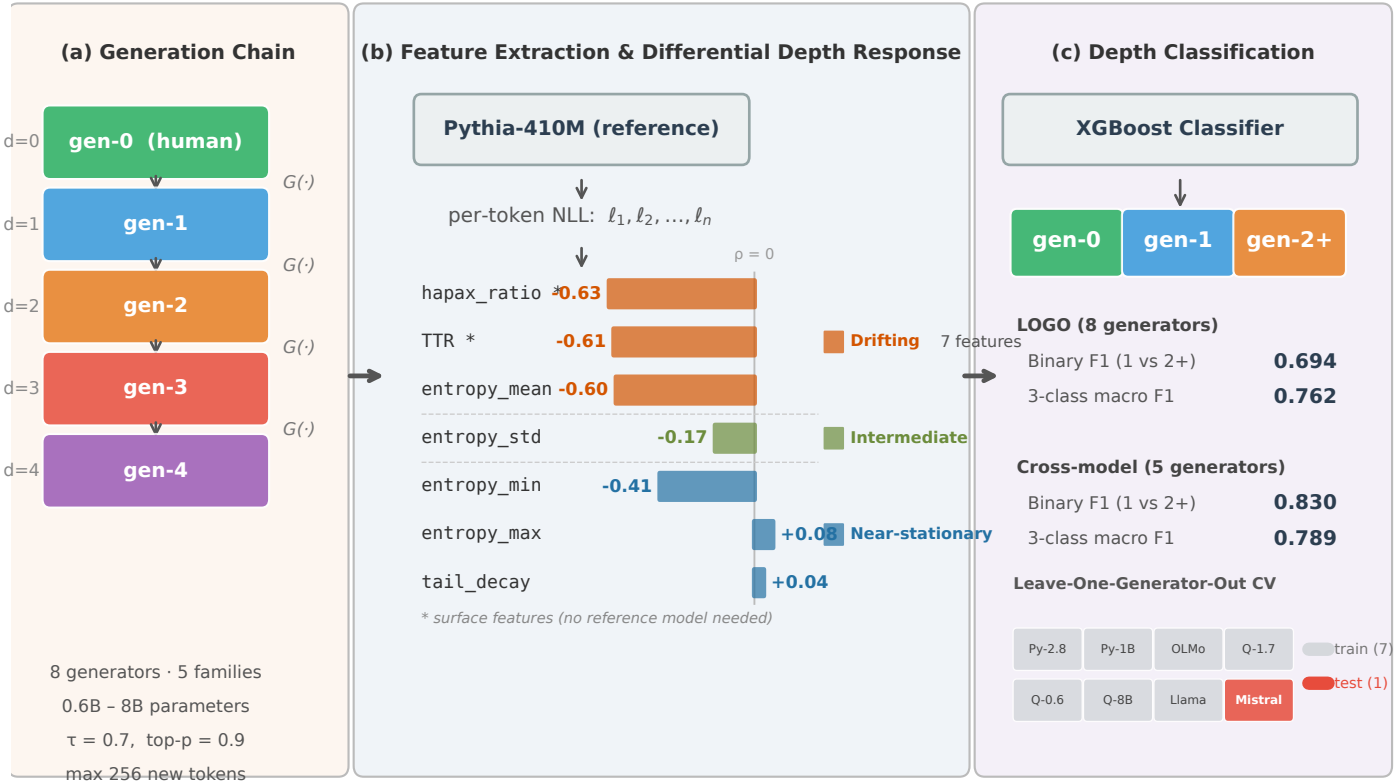
(a) Generation Chain
d=0
gen-0 (human)
G(·)
d=1
gen-1
G(·)
d=2
gen-2
G(·)
d=3
gen-3
G(·)
d=4
gen-4
8 generators · 5 families
0.6B – 8B parameters
τ = 0.7, top-p = 0.9
max 256 new tokens
(b) Feature Extraction & Differential Depth Response
Pythia-410M (reference)
per-token NLL: $\ell_1, \ell_2, \ldots, \ell_n$
ρ = 0
hapax_ratio * -0.63
TTR * -0.61
entropy_mean -0.60
entropy_std -0.17
entropy_min -0.41
entropy_max +0.08
tail_decay +0.04
Drifting
Intermediate
Near-stationary
7 features
* surface features (no reference model needed)
(c) Depth Classification
XGBoost Classifier
gen-0
gen-1
gen-2+
LOGO (8 generators)
Binary F1 (1 vs 2+) 0.694
3-class macro F1 0.762
Cross-model (5 generators)
Binary F1 (1 vs 2+) 0.830
3-class macro F1 0.789
Leave-One-Generator-Out CV
Py-2.8
Py-1B
OLMo
Q-1.7
Q-0.6
Q-8B
Llama
Mistral
train (7)
test (1)